Ananya Saha

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ACADEMIC DETAILS

Indian Institute of Technology (BHU) Varanasi
B.Tech and M.Tech in Civil Engineering (2022-2027)

- Major CGPA 8.22 out of 10.00
- Relevant courses: Traffic Engineering, Advanced Transportation Engineering, Urban Transport Systems Planning, Analysis of Transportation Systems, Modelling Analysis Simulation, Probability and Statistics

RESEARCH INTERESTS

Multimodal accessibility, urban freight, last-mile logistics, spatial and network analysis, transport equity and safety, travel behaviour, public transit networks, demand modelling, urban data quality and uncertainty, network resilience and disruption recovery, large-scale mobility data analytics, transport policy and infrastructure planning.

TECHNICAL SKILLS

Programming	Python (NumPy, pandas, GeoPandas, Shapely, NetworkX, rasterio, scikit-learn, matplotlib), R, MATLAB, MySQL, MS Excel, Git/GitHub, LaTeX
Geospatial	QGIS, OpenStreetMap (Overpass, JOSM), network analysis and routing, VIIRS NTL, GHSL
Statistics	Regression (OLS, logistic, Poisson), factor analysis, spatial statistics and econometrics, clustering (HDBSCAN, GMM), spatial null models, time-series analysis, sensitivity analysis
Simulation	PTV Vissim (COM interface), SUMO, NetLogo, CARLA, ROS2, Roboflow

PUBLICATIONS & PRESENTATIONS

1. **Saha, A.**, Mishra, S., & Pani, A. (2027). Multimodal Freight Accessibility and the Limits of Spatial Planning for Locational Disadvantage: Case Studies of Manufacturing Enterprises in Five Indian Cities. *Transportation Research Board (TRB) 106th Annual Meeting, Washington, D.C., Jan 2027.* Accepted for presentation. [Code]

2. Kumar, N., Mishra, S., & **Saha, A.** (2027). Infrastructure-Demand Decoupling along India's Eastern Dedicated Freight Corridor: Freight Desert Mapping and Spatial Regression Evidence from 2.9 Million Establishments *Transportation Research Board (TRB) 106th Annual Meeting, Washington, D.C., Jan 2027.* Accepted for presentation.

3. Mishra, S., **Saha, A.**, Pani, A. Does port governance reach the hinterland? Terminal governance, manufacturing location and port-hinterland alignment in five Indian cities. *Maritime Policy & Management*. Under review.

PROJECTS & RESEARCH EXPERIENCE

1. **Multimodal Freight Accessibility and the Limits of Spatial Targeting for Locational Disadvantage**
Dept. of Civil Engineering, IIT (BHU) Varanasi (Mar – Jul 2026) Guide: Dr. Agnivesh Pani [Code]

   - Built a multimodal freight accessibility index for 123,591 manufacturing MSMEs in five Indian cities from Dijkstra distances to road, rail and ICD terminals with externally anchored impedances.
   - Tested robustness across a 625-combination sensitivity sweep; identified freight-deprivation zones with HDBSCAN and tested their spatial targetability against spatial null models.
   - Found freight disadvantage rarely forms compact zones, favouring network over area-based fixes.

2. **Trajectory-Calibrated Microsimulation of Highway Work-Zone VMS Compliance under Mixed Traffic**
IIT Hyderabad – Summer Research Internship (May – Jul 2026) Guide: Dr. Digvijay Pawar

- Built a calibrated Vissim model of a National Highway work-zone lane closure in mixed traffic, capturing merging, speed adaptation and lane changing upstream of the taper.
- Modelled Variable Message Sign (VMS) messaging across driver-compliance rates and sign distances, measuring effects on merge timing, delay, queue length and throughput.
- Developed a reusable framework for testing VMS and work-zone strategies in heterogeneous traffic.

3. **Work Patterns, Fatigue, and Safety Events among E-Hailing Drivers: A Cross-Sectional Study**
Universiti Teknologi Malaysia (UTM) – Research Project (Jan – May 2026) Guide: Dr. Nordiana Mashros

- Modelled how work patterns and demographics shape fatigue and safety-critical events (e.g. dozing at the wheel) among 363 drivers via hierarchical OLS, staged logistic/Poisson regression and CFA.
- Part-time driving raised fatigue (Cohen's d = 0.755); ages 35–44 had higher dozing odds (OR = 3.531).

4. **Queue-Length Based Adaptive Signal Control at a Heterogeneous Roundabout (VISSIM COM)**
Dept. of Civil Engineering, IIT (BHU) Varanasi (Aug 2025 – Jan 2026) Guide: Dr. Abhisek Mudgal

- Implemented a queue-length based adaptive controller via VISSIM COM at a congested heterogeneous roundabout, allocating green time in proportion to smoothed stage queues.
- Compared delay, queue length and throughput with fixed-time control under encroachment, off-peak, lane-closure and rainfall scenarios.

5. **Development of a 1:14 Scaled Connected Autonomous Vehicle Testbed for Mixed Traffic Conditions**
IIT Bombay – Summer Research Internship (May – Jul 2025) Guide: Dr. Archak Mittal

- Designed a 1:14 cyber-physical testbed (with RWTH Aachen) for non-lane-based Indian traffic, integrating CARLA, Unreal Engine 5, TrueVision, RoadRunner and CommonRoad.
- Built a pipeline bringing real-world maps into CARLA, computing geometry, turning radii, clearances, speeds, lane widths and scaling; validated Lanelet2 topology in JOSM and ROS2 (RViz).
- Wrote a multi-agent CARLA script to spawn, control and monitor vehicles with a live top-down view.

6. **Pedestrian Crossing Behaviour Study at Mid-Block Sections**
Dept. of Civil Engineering, IIT (BHU) Varanasi (Feb – May 2025) Guide: Dr. Abhisek Mudgal

- Recorded mid-block crossings in mixed traffic with little or no signal control (video and manual observation), measuring gap acceptance, delay, crossing speed and vehicle conflicts.
- Modelled how age, group size, vehicle density and visibility shape crossing decisions; identified unsafe behaviours and quantified pedestrian Level of Risk to inform design and traffic calming.

7. **SafeWalk Kolkata: Night-Time Walking Exposure and Route Choice**
Independent project (Sep – Nov 2026) [Code] [Live map]

- Scored 74,288 street segments (4,434 km) for night-time exposure from OSM lighting, opening hours and land use plus VIIRS night lights, with per-segment confidence separating observation from prior.
- Validated without ground truth: safer routes halve exposed walking for a 1.2% median detour (200 trips); rankings are robust to weights (Spearman $\rho \geq 0.976$); data confidence tracks night-time radiance ($\rho = 0.63$), so data is thinnest in darker areas.
- Built a serverless browser route planner with hourly scores, metro last-mile and search.

ACHIEVEMENTS & LEADERSHIP

- Ranked in the top 10 percentile (among a million candidates) in JEE Mains.
- Graduate researcher at SCULPT Lab (PI: Dr. Agnivesh Pani), IIT (BHU) Varanasi.
- Member of Students' Parliament and Entrepreneurship Cell, IIT (BHU) Varanasi.
- Runner-up in Civil Conclave Case Study, IIT Roorkee '24